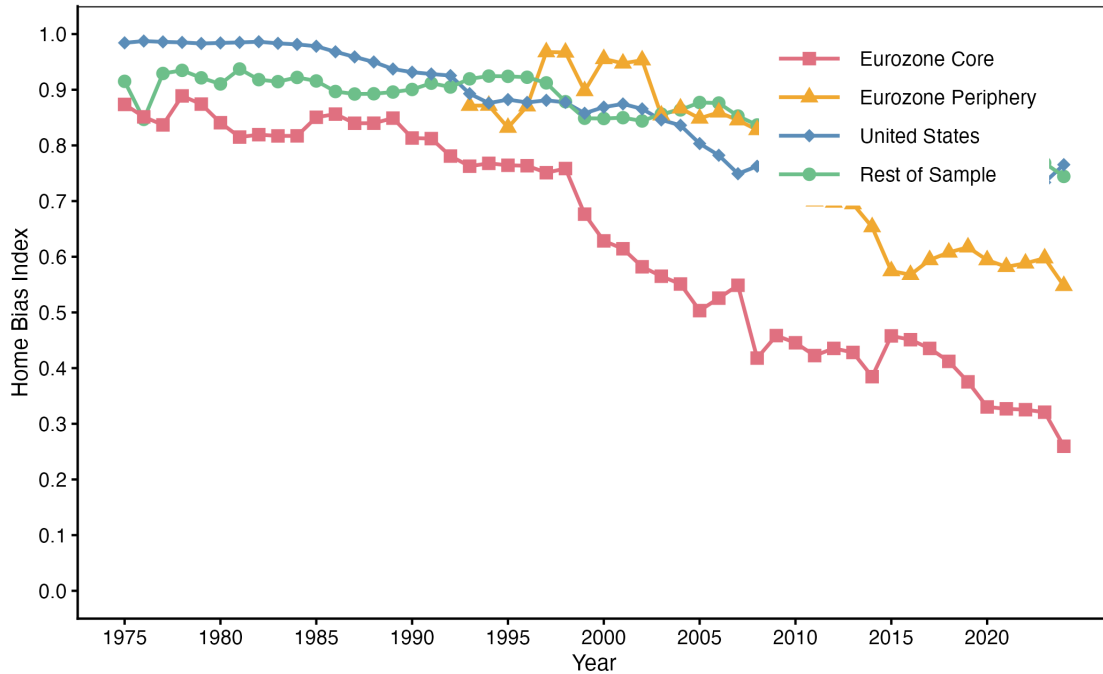


ECB Paper Extension Financial Hubs and Home Bias in the EU

May 24, 2026

1 Equity Home Bias and Euro Area Financial Integration

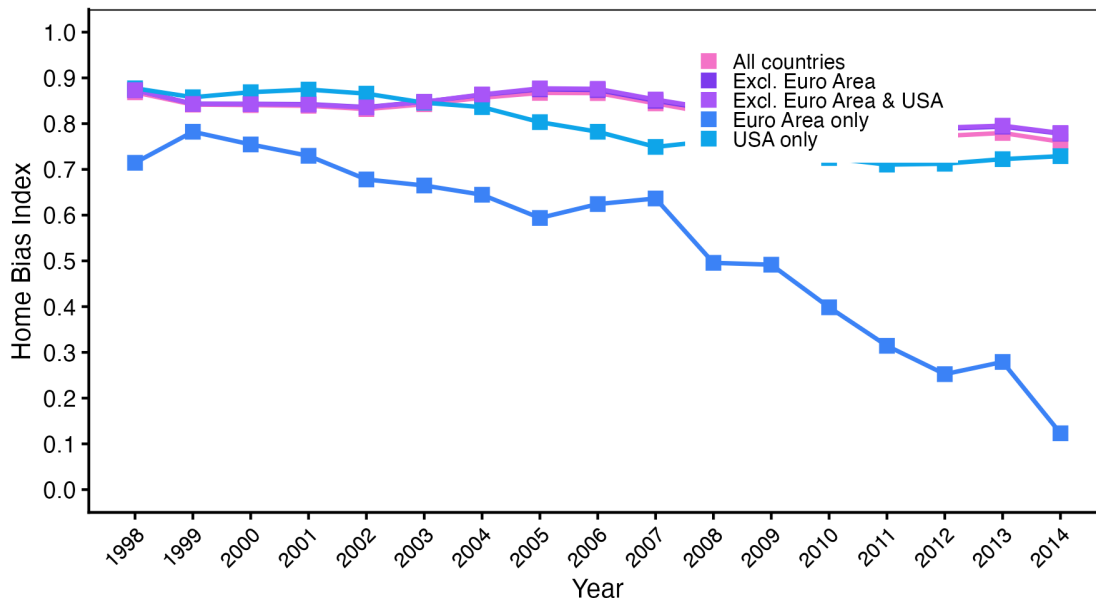
Our analysis of the EWN data revealed two striking patterns. First, Euro Area countries exhibited a markedly stronger decline in equity home bias than non-Euro Area OECD countries over the sample period (Figure ??). This is consistent with the hypothesis that the introduction of the common currency facilitated financial integration among member states, reducing barriers to cross-border equity investment within the currency union. Second, and more puzzling, the Euro Area as a consolidated entity also exhibited a pronounced EHB decline approaching 0, and thus substantially exceeding that of comparable non-EA developed economies.



Note: Unweighted cross-country mean equity home bias index by country group. Eurozone Core = founding 1999 members; Eurozon

Figure 1: Equity home bias by country group, 1975–2024.

The second finding is difficult to reconcile based on what we would expect from European Integration. If the decline in individual countries' home bias were primarily driven by increased holdings of other Euro Area countries' equities (intra-EA integration), then consolidating the Euro Area into a single entity should reclassify these cross-border intra-EA positions as domestic, leaving the aggregate EHB largely unchanged relative to other developed economies. The fact that the aggregate declined sharply suggests either that EA countries also diversified substantially into non-EA equities, or that the underlying data suffers from measurement problems that inflate measured foreign equity positions.



Note: Unweighted cross-sectional means.

Figure 2: Equity home bias for the Euro Area aggregate, USA and rest of sample.

1.1 Measurement Distortions in Euro Area Portfolio Data

A recent contribution by [?] provides a resolution to this puzzle. The authors demonstrate that standard international portfolio statistics—the International Investment Position (IIP) and the IMF’s Coordinated Portfolio Investment Survey (CPIS)—systematically overstate the foreign equity positions of Euro Area countries due to the role played by what they term “onshore offshore financial centers” (OOFCS): Luxembourg, Ireland, and to a lesser extent the Netherlands. These countries serve two functions that distort measured bilateral positions.

Fund intermediation. Luxembourg and Ireland host large investment fund industries. These funds are domiciled there for regulatory and tax reasons (the UCITS passport, favorable withholding tax treaties), but their investors reside worldwide. Under standard residency-based accounting, when a German pension fund purchases shares in a Luxembourg-domiciled equity fund, this is recorded as Germany holding a foreign equity claim on Luxembourg—regardless of what the fund actually invests in. The underlying exposures (to US, Japanese, or even German equities) are invisible in the data. This creates two problems. First, it misattributes the geographic destination of investment: a claim that economically represents exposure to the US equity market appears as exposure to Luxembourg. Second, and more consequentially, a substantial share of the assets managed by Luxembourg and Ireland funds is owned by non-EA investors—the United Kingdom, United States, Switzerland, Japan, and others. These positions are attributed to the Euro Area in standard statistics because the fund is domiciled within the EA, even though the ultimate investors are not European.

Securities issuance by foreign affiliates. Multinational corporations routinely establish financing subsidiaries in the Netherlands, Ireland, or Luxembourg for tax and regulatory purposes. When an Italian investor purchases a bond issued by such a subsidiary, the position is recorded as a claim on the subsidiary’s country of residence rather than the parent company’s home country. This channel is addressed by a nationality-based restatement of securities, which we do not pursue further here but which [?] document in detail.

Three portfolio components. The key conceptual contribution of [?] is to decompose measured Euro Area portfolios into three components. The first comprises direct holdings of securities by EA investors—the undistorted portion of the data. The second consists of indirect holdings by EA investors through OOFD-domiciled funds: when these funds are “unwound,” the fund share claim on Luxembourg or Ireland is replaced by the underlying securities the fund actually holds. The third, and most problematic, component consists of positions held by rest-of-world investors through OOFD funds. These are spuriously attributed to the Euro Area under residency-based accounting.

The authors estimate that of approximately €7 trillion in equities and bonds held by Luxembourg and Ireland funds, only around €2.4 trillion can be attributed to EA investors. The remaining €4.7 trillion belongs to the rest of the world but inflates measured EA positions in standard statistics. For equity home bias specifically, [?] show that after correcting for these distortions, the apparent excess decline in EA equity home bias relative to other developed economies largely disappears (their Figure 9). The corrected EA average EHB follows a trajectory comparable to that of the United States and other non-EA developed economies.

1.2 Who Owns Luxembourg and Ireland Funds?

To illustrate the magnitude of the attribution problem, we use the fund counterparty statistics compiled by [?] from regulatory data reported by the Luxembourg Commission de Surveillance du Secteur Financier (CSSF) and the Central Bank of Ireland. These data record the immediate counterparty—the investor who directly holds fund shares—by country or region of residence.

Figure ?? shows the composition of ownership of Luxembourg-domiciled fund shares from 1998 to 2021. Throughout the entire period, Euro Area investors (Luxembourg residents and the rest of the EA combined) account for less than half of total fund assets. The remaining share is held by investors from Switzerland, the United Kingdom, the United States, Japan, and other non-EA countries. In standard portfolio statistics, the entirety of these funds’ asset holdings is attributed to the Euro Area. The figure also reveals that total assets under management grew from approximately €460 billion in 1998 to over €5 trillion by 2021 (Figure ??), implying that the magnitude of the misattribution has increased substantially over time—precisely the period during which Euro Area equity home bias appeared to decline most rapidly.

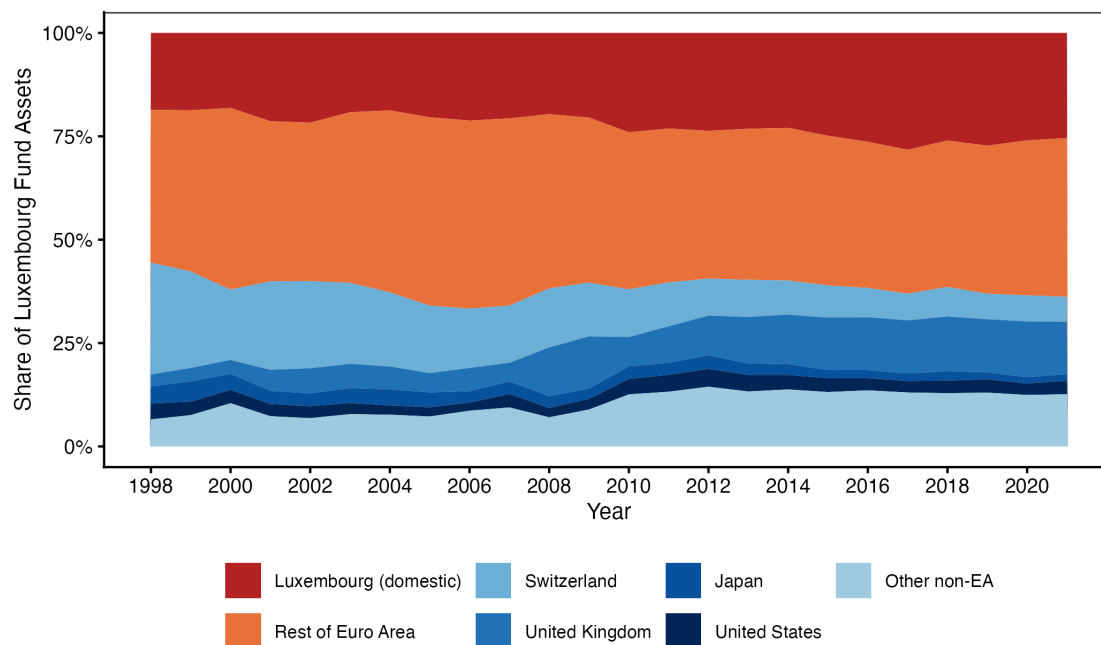
Figure ?? extends the analysis to Ireland using country-level counterparty data available for 2014–2021. The non-EA share is even more pronounced for Irish funds, where the United Kingdom alone accounts for a large fraction of total holdings. This is consistent with the role of Irish-domiciled funds as vehicles for UK-based institutional investors, including liability-driven investment (LDI) funds that [?] document in detail. The contrast between the two panels underscores that the misattribution problem, while present in both centers, operates through somewhat different investor bases.

1.3 Geographic Reallocation of Equity Positions

The fund counterparty statistics establish that a large share of OOFD fund assets belongs to non-EA investors. We now turn to the question of how this distorts the measured bilateral equity positions of individual Euro Area countries. Using the SHS-based restated bilateral portfolio data from [?], which covers all EA countries as investors against approximately 240 destination countries for the period 2014–2023, we compare residency-based positions with fund-unwind-adjusted positions for selected EA economies.

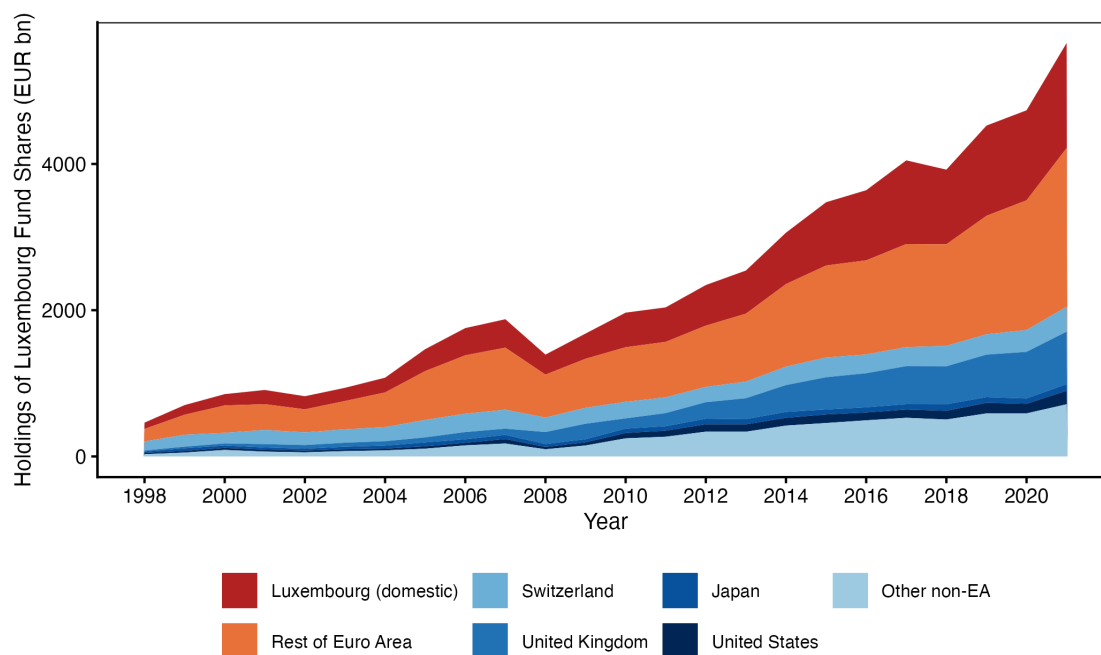
Figure ?? shows Germany’s top foreign equity destinations in 2020 under both methodologies. The contrast is stark. Under residency-based accounting, Luxembourg and Ireland rank among Germany’s largest foreign equity positions, reflecting German investors’ substantial holdings of OOFD-domiciled fund shares. After the fund unwind—which replaces each fund share with the underlying securities the fund holds—positions on Luxembourg and Ireland collapse, while positions on the United States and other major equity markets expand correspondingly. The pattern is similar for France (Figure ??) and particularly pronounced for Belgium (Figure ??), whose geographic proximity to Luxembourg is reflected in an especially large share of fund-intermediated investment.

These paired bar charts show where positions are reallocated *to*, but do not directly reveal the extent of round-tripping—the phenomenon whereby an investor’s holdings of OOFD fund shares partly represent



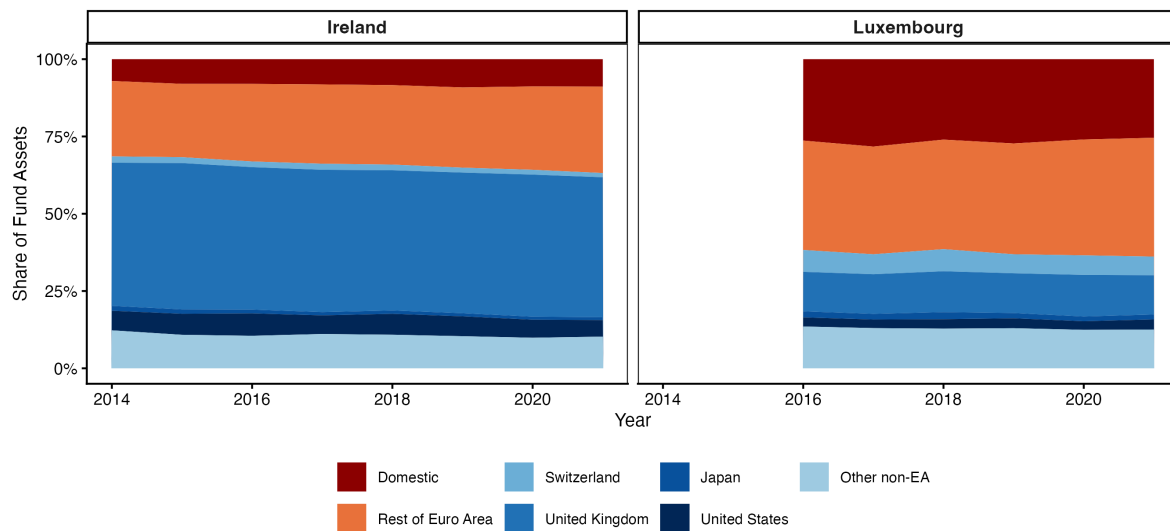
Note: Ownership shares of Luxembourg-domiciled fund shares by immediate counterparty region.
Source: CSSF via Beck et al. (2024).

Figure 3: Ownership composition of Luxembourg-domiciled fund shares by immediate counterparty region, 1998–2021. Euro Area investors (domestic Luxembourg and rest of EA) consistently account for less than half of total assets.



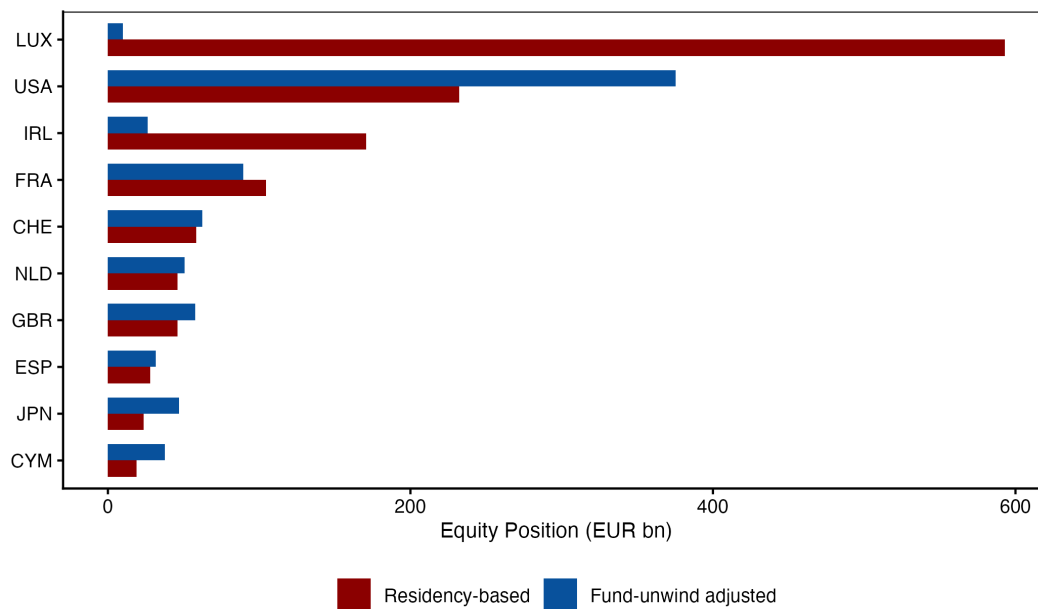
Note: Holdings of Luxembourg-domiciled fund shares by immediate counterparty region.
Source: CSSF via Beck et al. (2024).

Figure 4: Holdings of Luxembourg-domiciled fund shares in EUR billions by immediate counterparty region, 1998–2021.



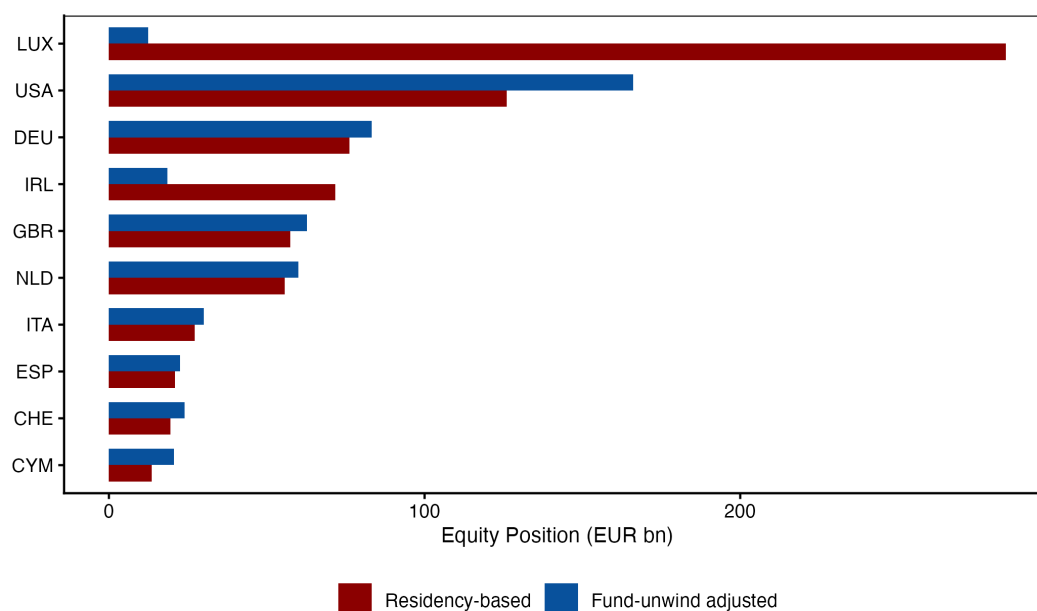
Note: Ownership shares by immediate counterparty. 'Domestic' = Irish investors for Ireland funds, Luxembourg investors for Luxembourg funds.
Source: Central Bank of Ireland and CSSF via Beck et al. (2024).

Figure 5: Ownership composition of Irish and Luxembourg fund shares by immediate counterparty country, 2014–2021.



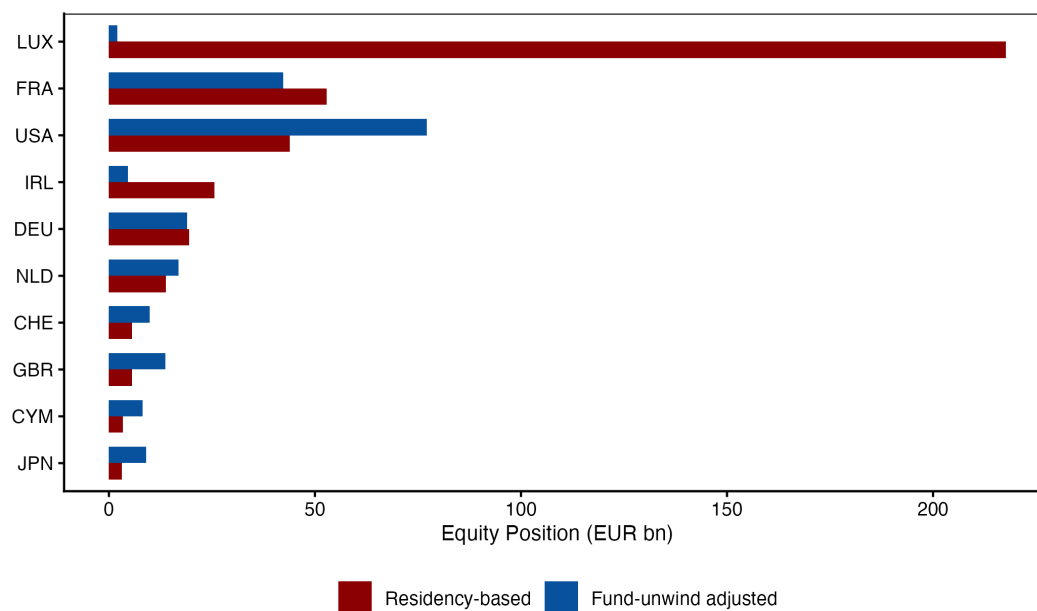
Note: Top 10 foreign equity destinations for DEU (2020q4). Residency-based vs. fund-unwind adjusted.
Source: Beck et al. (2024), SHS-based restated bilateral portfolios.

Figure 6: Germany's top foreign equity destinations: residency-based vs. fund-unwind adjusted positions, 2020Q4.



Note: Top 10 foreign equity destinations for FRA (2020q4). Residency-based vs. fund-unwind adjusted.
Source: Beck et al. (2024), SHS-based restated bilateral portfolios.

Figure 7: France's top foreign equity destinations: residency-based vs. fund-unwind adjusted positions, 2020Q4.



Note: Top 10 foreign equity destinations for BEL (2020q4). Residency-based vs. fund-unwind adjusted.
Source: Beck et al. (2024), SHS-based restated bilateral portfolios.

Figure 8: Belgium's top foreign equity destinations: residency-based vs. fund-unwind adjusted positions, 2020Q4.

indirect claims on domestic equities. To capture this, Figure ?? displays the net reallocation for each destination, defined as the difference between the fund-unwind-adjusted and residency-based position. Destinations with negative values (notably Luxembourg and Ireland) are those from which positions are reallocated away; destinations with positive values are those that gain exposure after the correction. The domestic position—labeled with the investor’s own country code—captures the round-tripping effect: equity that appeared as a foreign claim on Luxembourg but is in fact an indirect holding of domestic securities. While the round-tripping component is non-negligible, it is small relative to the geographic reallocation toward the United States and other major markets, indicating that the primary distortion operates through misattribution of destination rather than through artificial inflation of the foreign portfolio share.

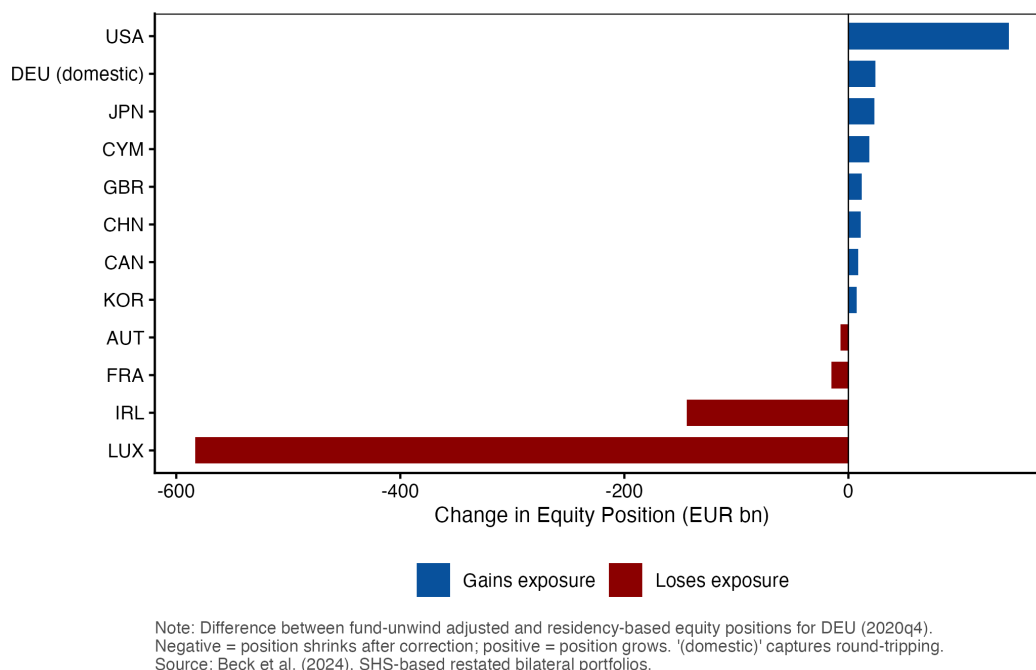
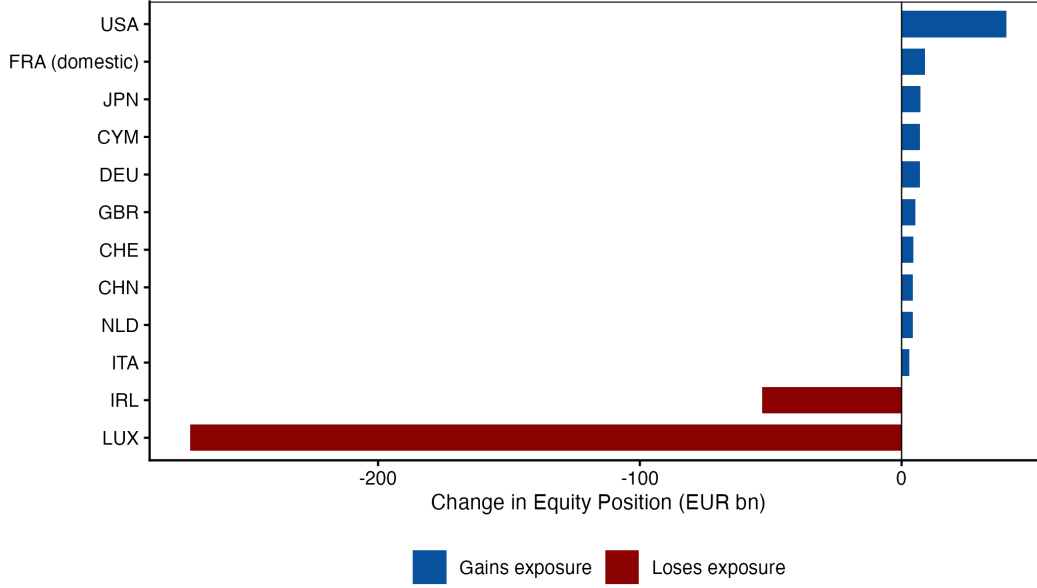


Figure 9: Net reallocation of Germany’s equity positions after fund unwind, 2020Q4. Negative values indicate positions that shrink; positive values indicate positions that grow. The domestic bar captures round-tripping through OOFC funds.

1.4 Implications for EWN-Based Home Bias Measures

Having established the mechanisms through which OOFC intermediation distorts measured bilateral positions, we can now assess how these distortions affect the two EWN-based equity home bias measures presented in Figures ?? and ??: the average EHB across individual Euro Area countries and the EHB of the Euro Area treated as an aggregate.

Individual EA country EHB. For the equity home bias of an individual country, only the *total* foreign equity share matters, not its geographic composition. The reallocation of positions from Luxembourg to the United States or Japan, while substantial in bilateral terms, does not affect a country’s measured foreign portfolio share: both the misattributed and the corrected positions count as foreign. The only channel through which the fund intermediation distortion biases individual countries’ EHB is round-tripping—the portion of OOFC fund holdings that represents indirect claims on domestic equities. When a German investor holds a Luxembourg fund that in turn holds German equities, this position is recorded as a foreign claim on Luxembourg rather than a domestic holding. Foreign equity assets are therefore overstated, domestic holdings understated, and the resulting EHB is biased downward. As Figure ?? illustrates, this round-tripping component exists but is modest relative to the total portfolio. The downward bias in individual EA



Note: Difference between fund-unwind adjusted and residency-based equity positions for FRA (2020q4). Negative = position shrinks after correction; positive = position grows. 'domestic' captures round-tripping. Source: Beck et al. (2024), SHS-based restated bilateral portfolios.

Figure 10: Net reallocation of France’s equity positions after fund unwind, 2020Q4. Negative values indicate positions that shrink; positive values indicate positions that grow. The domestic bar captures round-tripping through OOFC funds.

countries’ EHB from the EWN data is therefore present but limited in magnitude, and our finding that EA countries experienced a stronger EHB decline than non-EA OECD countries is unlikely to be entirely driven by this distortion—though it may be somewhat exaggerated. The country average we used excluded Ireland and Luxembourg which EHB is very distorted.

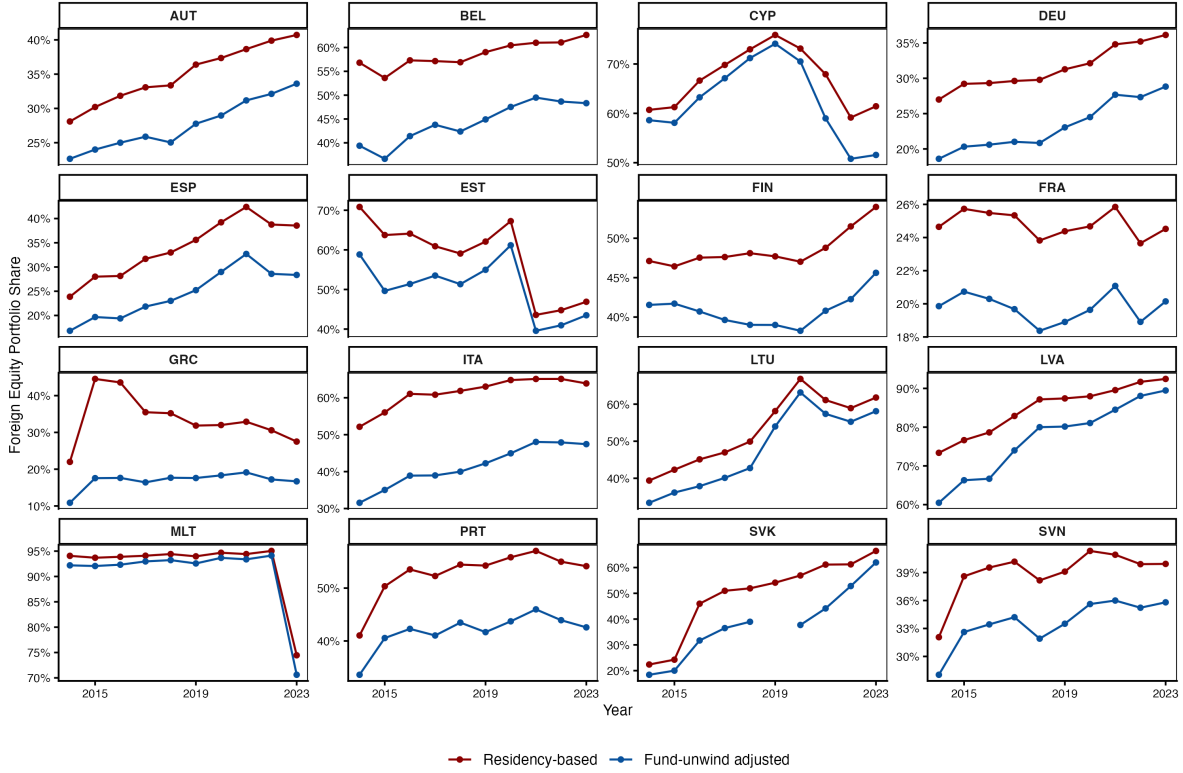
Euro Area aggregate EHB. The distortion operates through an entirely different channel at the aggregate level. When the Euro Area is treated as a single entity, intra-EA cross-border holdings are reclassified as domestic. This consolidation eliminates both the round-tripping problem (indirect domestic claims via OOFC funds are domestic from the aggregate perspective regardless) and the intra-EA geographic misattribution (a German claim recorded against Luxembourg but actually representing French equity is intra-EA in all cases). The distortion that *does not* cancel at the aggregate level is the presence of rest-of-world investors in OOFC-domiciled funds. When a US or UK investor holds shares in a Luxembourg fund that invests in non-EA equities, the fund’s assets are recorded as Luxembourg’s—and therefore the Euro Area’s—foreign equity claims on the rest of the world. In reality, this represents rest-of-world capital investing in rest-of-world equities, with Luxembourg serving merely as an intermediary. This channel inflates the Euro Area aggregate’s measured foreign equity position and biases its EHB downward. As the fund counterparty statistics in Figure ?? demonstrate, non-EA investors account for more than half of Luxembourg fund assets throughout the sample period, and the absolute magnitude of this misattribution has grown from hundreds of billions to several trillion euros—explaining why the Euro Area aggregate EHB declined so dramatically in the EWN data despite the consolidation of intra-EA positions.

1.5 Corrected Foreign Equity Portfolio Shares

To quantify the distortion directly, we use the SHS-based restated bilateral portfolio data to compute the foreign equity portfolio share $\omega_{j,t} = \text{ForeignEquity}_{j,t} / (\text{ForeignEquity}_{j,t} + \text{DomesticEquity}_{j,t})$ for each EA country under both the residency-based and the fund-unwind-adjusted methodology. This share constitutes the key input to the equity home bias measure: the EHB is a monotone decreasing function of ω for a

given market capitalization weight. While we cannot compute the full EHB from the SHS data alone (which would require reliable market capitalization data that is unavailable for recent years), comparing ω across methodologies reveals the direction and magnitude of the bias.

Figure ?? displays the two series for each EA country (excluding OOFCS) over 2014–2023. The gap between the residency-based and fund-unwind-adjusted foreign share varies substantially across countries, with Belgium exhibiting the largest distortion and smaller or newer EA members showing more modest differences. This heterogeneity reflects differences in the extent to which each country’s investors rely on OOFCS-domiciled funds.



Note: Foreign share = foreign equity / (foreign + domestic equity). EA countries excluding Luxembourg, Ireland, and the Netherlands.
Source: Beck et al. (2024), SHS-based restated bilateral portfolios.

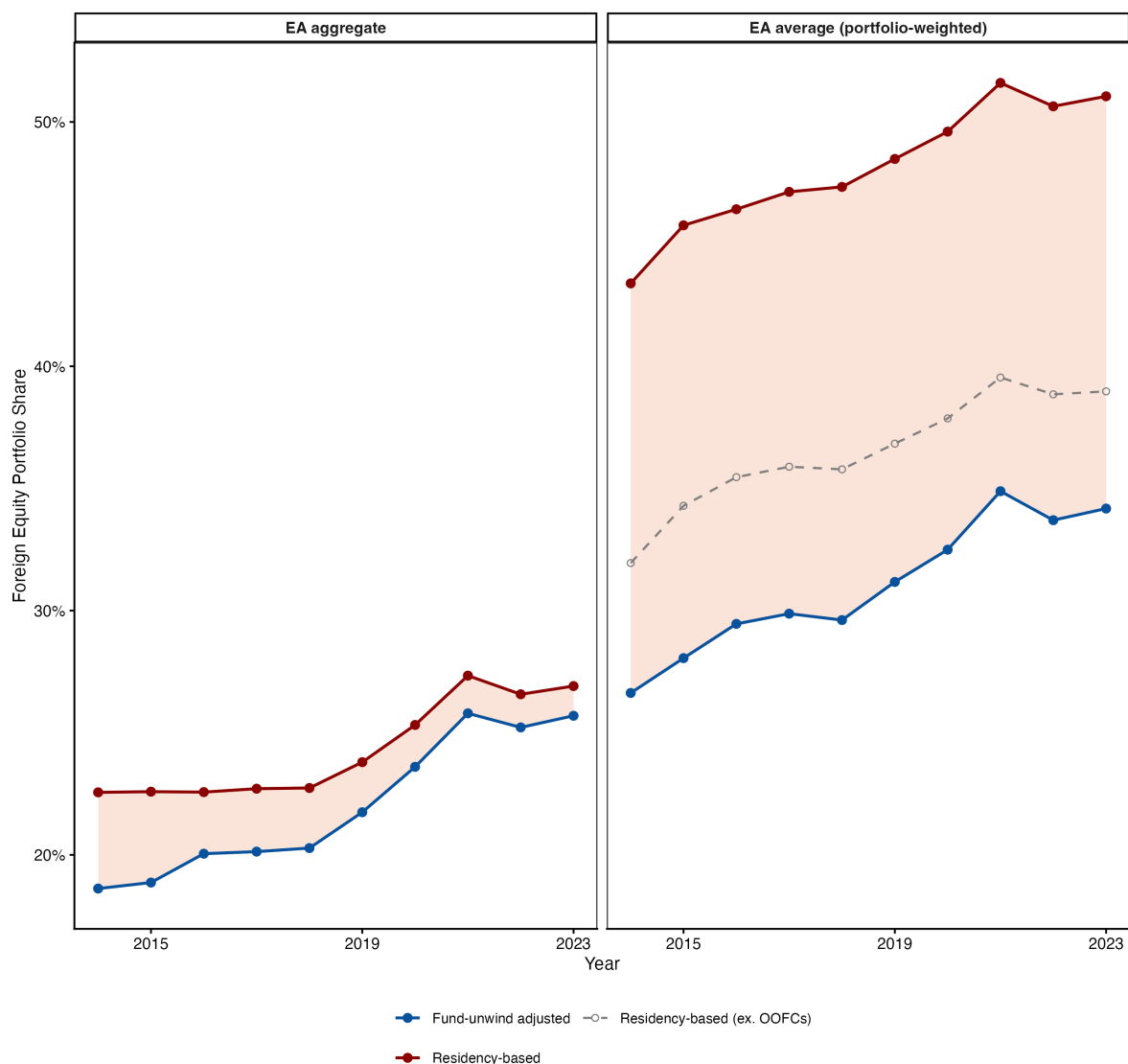
Figure 11: Foreign equity portfolio share by EA country, 2014–2023. Residency-based vs. fund-unwind adjusted. EA countries excluding Luxembourg, Ireland, and the Netherlands.

Figure ?? presents the central comparison of this analysis. The left panel shows the foreign equity portfolio share of the Euro Area treated as a single entity, where intra-EA cross-border holdings are classified as domestic. The right panel shows the portfolio-weighted average across individual EA countries, including the OOFCS. In both panels, the shaded area represents the overstatement of foreign equity exposure attributable to fund intermediation.

Two features are noteworthy. First, the level of the corrected foreign share (blue line) is substantially lower in the aggregate panel than in the country average panel—approximately 26% versus 34% in 2023. This difference reflects genuine intra-EA cross-border equity holdings that count as foreign at the country level but become domestic at the aggregate level, consistent with financial integration within the currency union.

Second, the overstatement from fund intermediation operates through different channels in the two panels. For the country average, the gap between the residency-based and corrected series is driven primarily by the inclusion of OOFCS in the weighted average: under residency-based accounting, Luxembourg and Ireland carry disproportionate portfolio weights and near-complete foreign shares, inflating the average. The dashed grey line, which shows the residency-based average excluding OOFCS, confirms that the distortion affecting

non-OOFC countries individually is more limited. For the Euro Area aggregate, the overstatement arises from a different source: non-EA investors' holdings of OOFC fund shares that invest in non-EA equities, which are spuriously counted as the Euro Area's foreign claims on the rest of the world.



Note: Left panel — EA treated as single entity (intra-EA = domestic). Right panel — portfolio-weighted average of individual EA countries. Dashed grey line shows residency-based average excluding Luxembourg, Ireland, and the Netherlands. Source: Beck et al. (2024), SHS-based restated bilateral portfolios.

Figure 12: Foreign equity portfolio share: EA aggregate vs. portfolio-weighted EA country average, 2014–2023. Shaded areas show overstatement from fund intermediation. Dashed grey line shows residency-based average excluding OOFCs.

1.6 Revisiting the Hypothesis

We return to the puzzle motivating this analysis. Our initial hypothesis was that the introduction of the euro facilitated financial integration among member states, reducing equity home bias primarily through increased intra-EA cross-border investment. This implied that while individual EA countries' EHB should decline more rapidly than that of non-EA economies, the Euro Area aggregate EHB—where intra-EA positions are consolidated as domestic—should not. The EWN data appeared to contradict this prediction, showing a

strong decline in the aggregate as well.

The analysis presented in this section resolves this tension. The apparently anomalous decline in the Euro Area aggregate EHB is substantially driven by the spurious attribution of rest-of-world investors' fund holdings to the Euro Area. After correcting for fund intermediation, the overstatement of the aggregate's foreign equity share is reduced, as shown in the left panel of Figure ?? . At the same time, the strong decline observed for individual EA countries in the EWN data is partly—though not entirely—an artifact: the round-tripping channel biases individual countries' EHB downward, and any portfolio-weighted average that includes the OOFs is further inflated by their extreme residency-based values.

The corrected data does lend some support to the integration hypothesis, though in a more nuanced form than originally proposed. The level difference between the corrected foreign equity shares in the two panels of Figure ??—approximately 8 percentage points in 2023—represents genuine intra-EA equity holdings that are foreign from individual countries' perspective but domestic from the aggregate's. Moreover, this level difference implies a systematically higher EHB for the Euro Area aggregate than for the average of its member states, since the home bias measure $EHB = 1 - \omega / (1 - A)$ is affected by both a lower foreign share ω at the aggregate level and a larger market capitalization weight A in the denominator: the Euro Area as a whole constitutes a substantially larger share of world equity market capitalization than any individual member state, so $(1 - A_{EA})$ is smaller than $(1 - A_j)$ for any country j . Both effects push the aggregate EHB above the country average, consistent with what intra-area integration would predict.

However, two important caveats apply. First, the SHS-based corrected data is only available from 2014 onward, well after the introduction of the euro in 1999. The corrected foreign equity shares in both panels follow roughly parallel upward trends over 2014–2023, offering no direct evidence that the level gap between the aggregate and the country average widened as a consequence of monetary integration. The observed level difference is equally consistent with gravity-based explanations: countries that are geographically and culturally proximate tend to invest more in each other's equity markets regardless of currency arrangements. Without corrected data from the pre-euro period, we cannot disentangle the euro's causal contribution from a proximity effect that may have existed prior to 1999.

Second, [?] demonstrate that after applying the full set of corrections—fund unwind and nationality adjustment—to the standard equity home bias measure, the average EHB of Euro Area countries follows a trajectory comparable to that of the United States and other non-EA developed economies (their Figure 9). The apparent excess decline in EA equity home bias, which featured prominently in earlier assessments of European financial integration, largely disappears. In this sense, the euro does not appear to have generated an exceptional reduction in equity home bias relative to the global trend.

The picture is markedly different for bond markets. [?] show that the fund intermediation distortion primarily affects equity positions, because fund shares are classified as equity instruments under the Balance of Payments Manual (BPM6) accounting framework. Holdings of OOFs-domiciled fund shares therefore enter the equity portfolio but not the bond portfolio, regardless of the fund's underlying asset allocation. After applying their corrections to bond home bias, the authors find that the Euro Area still exhibits a genuine excess decline relative to other developed economies—from approximately 0.8 in the mid-1990s to around 0.35 by 2020, compared to a decline from 0.9 to 0.8 for the United States over the same period. This is consistent with the elimination of currency risk in bond markets being a particularly powerful driver of financial integration, as currency risk represents a more salient barrier for fixed-income investors than for equity investors. The hypothesis that the euro facilitated intra-area financial integration thus finds its strongest support not in equity markets, where measurement distortions cloud the picture, but in bond markets, where the data are less affected by OOF intermediation and the integration gains are unambiguous.

This asymmetry between debt and equity integration carries implications beyond the measurement debate. In the framework of [?], it is equity diversification—not debt—that drives income risk sharing through cross-border ownership of productive assets. The finding that the euro successfully integrated bond markets but left equity home bias at levels comparable to non-EA economies suggests that the currency union has delivered integration in precisely the asset class where the risk-sharing gains are smallest, while the channel that matters most for consumption smoothing remains underdeveloped. This resonates with ongoing policy discussions around the European Capital Markets Union and the diagnosis in the [?] report, which identifies the fragmentation of European equity and venture capital markets as a key barrier to innovation and growth. The persistent equity home bias within the Euro Area—even after two decades of monetary integration—underscores that a common currency alone is insufficient to overcome the structural, regulatory,

and informational barriers to cross-border equity investment, and that deeper capital market integration may require the kind of institutional reforms currently under discussion, including elements of a banking union and harmonized insolvency and securities regulation.